Accident Analysis and Prevention 157 (2021) 106157

Contents lists available at ScienceDirect

Accident Analysis and Prevention

journal homepage: www.elsevier.com/locate/aap

A review of surrogate safety measures and their applications in connected and automated vehicles safety modeling

Chen Wang [a,*], Yuanchang Xie [b], Helai Huang [c], Pan Liu [a]

[a] School of Transportation, Southeast University, China
[b] Department of Civil and Environmental Engineering, University of Massachusetts Lowell, 1 University Ave, Lowell, MA 01854, United States
[c] School of Traffic and Transportation Engineering, Central South University, China

ARTICLE INFO

Keywords:
Surrogate safety measure
Connected vehicles
Automated vehicles
Simulation
Safety

ABSTRACT

Surrogate Safety Measures (SSM) are important for safety performance evaluation, since crashes are rare events and historical crash data does not capture near crashes that are also critical for improving safety. This paper focuses on SSM and their applications, particularly in Connected and Automated Vehicles (CAV) safety modeling. It aims to provide a comprehensive and systematic review of significant SSM studies, identify limitations and opportunities for future SSM and CAV research, and assist researchers and practitioners with choosing the most appropriate SSM for safety studies. The behaviors of CAV can be very different from those of Human-Driven Vehicles (HDV). Even among CAV with different automation/connectivity levels, their behaviors are likely to differ. Also, the behaviors of HDV can change in response to the existence of CAV in mixed autonomy traffic. Simulation by far is the most viable solution to model CAV safety. However, it is questionable whether conventional SSM can be applied to modeling CAV safety based on simulation results due to the lack of sophisticated simulation tools that can accurately model CAV behaviors and SSM that can take CAV's powerful sensing and path prediction and planning capabilities into crash risk modeling, although some researchers suggested that proper simulation model calibration can be helpful to address these issues. A number of critical questions related to SSM for CAV safety research are also identified and discussed, including SSM for CAV trajectory optimization, SSM for individual vehicles and vehicle platoon, and CAV as a new data source for developing SSM.

1. Introduction

Surrogate Safety Measures (SSM) are important for traffic safety evaluation, largely due to the lack of reliable statistical safety models in many cases. This is particularly true for transportation facilities with complex site characteristics and/or nontraditional traffic safety treatments, where limited or no historical crash data is available for developing safety predictive models (Tarko et al., 2009). The research on SSM can date back to early 1970s (Hayward, 1971), and considerable achievements have been made in this field since then. Nowadays, SSM are increasingly being employed by researchers and practitioners to understand the safety implications of new traffic designs and various safety treatments using recorded videos and/or traffic simulation results.

An important application of SSM is modeling the safety impacts of Connected and Automated Vehicles (CAV) and their interactions with Human-Driven Vehicles (HDV). Research on CAV has attracted tremendous attention recently owing to their great potential to improve safety and traffic operations. However, the technology is not yet mature enough for large-scale CAV deployment in the real world. Therefore, CAV studies are mostly based on microscopic traffic simulations, and SSM have been widely adopted for quantifying the safety benefits of CAV based on simulation results. Many SSM have been proposed. This provides plenty of options to researchers and practitioners, but also presents challenges in terms of how to choose from them. Given the challenges and the abundant existing studies on SSM, now it is both an appropriate and important time to conduct a comprehensive and critical review of SSM, answering questions such as (1) what conventional SSM have been used in CAV safety studies; and (2) the pros and cons of these SSM. The answers will help researchers and practitioners choose the most appropriate SSM for various studies. Additionally, even though SSM have been popular for modeling new traffic safety designs and treatments and CAV, criticisms have been raised regarding their ability to accurately and objectively characterize a transportation facility's (e.

* Corresponding author.
E-mail addresses: wkobec@hotmail.com (C. Wang), yuanchang_xie@uml.edu (Y. Xie), huanghelai@csu.edu.cn (H. Huang), liupan@seu.edu.cn (P. Liu).

https://doi.org/10.1016/j.aap.2021.106157
Received 22 June 2020; Received in revised form 25 April 2021; Accepted 25 April 2021
Available online 8 May 2021
0001-4575/© 2021 Elsevier Ltd. All rights reserved.

g., intersection, road segment) safety performance. It is critical to review existing SSM, identify opportunities for improvements, and provide guidance for future research in this area.

1.1. Scope and aim

SSM has attracted substantial attention recently and some of the previous studies focusing on vulnerable road user safety have been nicely summarized by Johnsson et al. (2018). This article provides a comprehensive and critical review of SSM and discusses their various applications, especially in CAV related safety studies. The objectives of this study are three folds:

1 Understand the state-of-the-art of SSM and their applications in CAV safety studies;
2 Discuss the current issues of SSM used in CAV safety studies; and
3 Identify promising future research directions of SSM and their applications in CAV safety.

1.2. Structure

This paper is structured as follows: Section 2 provides a comprehensive review of the literature related to SSM. Section 3 discusses the pros and cons of SSM-based safety studies and reviews how SSM have been used in CAV safety evaluation. In Section 4, research questions and issues related to SSM for CAV safety studies are discussed. Finally, Section 5 concludes this study, discusses the main issues and gaps in SSM research particularly in the context of CAV, and identifies opportunities for further research.

2. Surrogate safety measures (SSM)

Over the past few decades, substantial efforts have been made in developing Surrogate Safety Measures (SSM). In this section, we start with providing an overview of SSM and CAV. Then, we summarize and discuss various SSM proposed in previous research.

2.1. Overview of SSM and CAV

2.1.1. SSM

Crash frequency and severity are considered two important indicators that directly measure the safety performance of a design, countermeasure, or system. However, crashes are rare events. For new safety strategies (e.g., a new traffic sign), it will take time for their safety impacts to be revealed by real-world crash frequency and severity data. Therefore, relying on historical crash data to evaluate a safety strategy's performance is not the best choice and to some extent is unethical. To address this issue, SSM derived from traffic conflicts have become an increasingly popular solution. Traffic conflicts are observable non-crash events, in which the interactions among multiple road users in space and time create a risk of collision if these users do not change their courses of movements (Amundsen and Hyden, 1977). A Conflict is considered to be etiologically connected to a crash, when a failure (e.g., human operator failure, road failure, and vehicle failure) that leads to the conflict cannot be properly corrected (Davis et al., 2011; Tarko, 2020). Due to the underlying causal relationship, measures used to identify traffic conflicts and quantify their severities could be considered as SSM. Compared to crashes, traffic conflicts are much more frequent.

It is worth noting that many safety related measures have been developed over time. However, not all of them can be considered as SSM. According to Tarko et al. (2009), two qualifying criteria for SSM are: (1) it should be derived from traffic conflicts which are directly linked to crashes, and (2) the relationship between traffic conflicts and the related potential crash frequency and/or severity can be quantified by some practical methods. From this perspective, traffic exposure/flow measurements, such as Annual Average Daily Traffic (AADT), speed variation, and average operating speed are not SSM, although these measurements have been proven to be associated with crash risk and are sometimes adopted as crash "surrogates". In this paper, only safety measures satisfying the above two qualifying criteria are reviewed and discussed.

2.1.2. CAV

Connected and Automated Vehicles (CAV) are interconnected and can perform driving functions with limited or completely without the assistance of human drivers. The level of automation ranges from Level 1 to Level 5, as shown in Table 1 (National Highway Traffic Safety Administration (NHTSA, 2016). At Levels 1 and 2, individual and combined assistant driving functions are automated, respectively. Level 3 allows a fully automated control under certain conditions. A full automation can be achieved at Levels 4 (only in certain circumstances) and 5. Currently, Level 2 automation technologies have been deployed in some automobiles and Level 3+ automation technologies are still under research and development. It can be expected that higher levels of automation will be achieved in the future.

CAV can communicate with drivers, other vehicles on the road (vehicle-to-vehicle [V2V]), roadside infrastructure (vehicle-to-infrastructure [V2I]), and the "Cloud" [V2C]. Through information sharing, early motion planning and automated control, CAV are expected to significantly improve traffic safety and mobility.

The behaviors of CAV can be very dissimilar from those of Human-Driven Vehicles (HDV) and their impacts are not well understood so far, since there are not many CAVs on the road yet and the technologies are evolving each day. Additionally, CAV at various automation and communication levels can behave differently. Both factors bring much complexity and uncertainty to CAV safety studies.

2.2. SSM and SSM-based models

In this section, we focus on reviewing important SSM and SSM-based models without going into details regarding their applications, which are discussed in Section 3. SSM are measures to identify traffic conflicts, which are statistically connected to crashes. The calculation of SSM in general is sensitive to the pre-determined thresholds used for defining traffic conflicts. After traffic conflicts are identified, SSM-based models are utilized to quantify the severity of a conflict. Some SSM-based models estimate the probability for a crash to occur, instead of

Table 1
Levels of Vehicle Automation (National Highway Traffic Safety Administration (NHTSA, 2016).

Levels of Automation	Who does what, when?
Level 0	The human driver does all the driving
Level 1	An advanced driver assistance system(ADAS) on the vehicle can sometimes assist the human driver with either steering or braking/accelerating, but not both simultaneously
Level 2	An advanced driver assistance system (ADAS) on the vehicle can itself actually control both steering and braking/ accelerating simultaneously under circumstances. The human driver must continue to pay full attention at all times and perform the rest of the driving task.
Level 3	An Automated Driving System (ADS) on the vehicle can itself perform all aspects of the driving task under some circumstances. In those circumstances, the human driver must be ready to take back control at any time when the ADS requires the human driver to do so. In all other circumstances, the human driver performs the driving task.
Level 4	An Automated Driving System (ADS) on the vehicle can itself perform all driving tasks and monitor the driving environment – essentially, do all the driving – in certain circumstances. The human need not pay attention in those circumstances.
Level 5	An Automated Driving System (ADS) on the vehicle can do all the driving in all circumstances. The human occupants are just passengers and need never be involved in driving.

generating a binary outcome (i.e., crash vs. non-crash).

2.2.1. SSM

As shown in Fig. 1, there are three main sub-categories under SSM, which are time-based SSM, deceleration-based SSM, and energy-based SSM. These three sub-categories of SSM are described in detail below. As explained later in this section, some distance-based SSM have also been proposed. Since these distance-based SSM rely on deceleration assumptions as well, they are covered in the deceleration-based SSM (Section 2.2.1.2) and are not discussed separately.

2.2.1.1. Time-based SSM. Time-based SSM measure the risk of an interaction in terms of its time proximity to a collision. The most common time-based SSM is Time-to-Collision (TTC), which was initially proposed by Hayward (Hayward, 1972) and defined as "*the time that remains until a crash between two vehicles would have occurred if the crash course and speed difference are maintained.*" An important assumption behind TTC is that both the speeds and directions of the involved vehicles are maintained. It is described mathematically as:

$$TTC_i(\mathrm{t}) = \frac{(X_{i-1}(t) - X_i) - l_i}{v_i(t) - v_{i-1}(t)} \tag{1}$$

Where,

X: vehicle position (i = following vehicle; $i-1$ = leading vehicle);
l: vehicle length; and
v: vehicle velocity.

Another popular time-based SSM is Time-to-Accident (TA) initially proposed by Perkins and Harris (1967). TA is calculated with estimated distance and speed when the evasive action is initially identified by a field observer. The major difference betweenTTC and TA is that TTC is measured at the beginning of the conflict occurrence, while TA is calculated based on the observed evasive action. Both indicators use certain thresholds to determine whether a conflict is at high risk or not.

Based on TTC, a few more complicated SSM were developed. Minderhoud and Bovy (2001) proposed Time-Exposed TTC (TET) and Time-Integrated TTC (TIT). TET is the time during a conflict when the TTC is below a certain threshold value, and TIT is the area between the threshold level and the TTC curve when the curve drops below the threshold. Compared to TTC and TA, TET and TIT focus on measuring the risk associated with the duration of dangerous driving conditions. Both TET and TIT require a continuous calculation of TTC.

$$TIT(t) = \sum_{n=1}^{N}\left[\frac{1}{TTC_{(it)}} - \frac{1}{TTC^*}\right]\Delta t,\ 0 < TTC(t) < TTC^* \tag{2}$$

$$TIT^* = \sum_{t=1}^{T} TIT(t) \tag{3}$$

$$TET(t) = \sum_{n=1}^{N} \delta_t \times \Delta t, \delta_t = \begin{cases} 1, & 0 < TTC(t) < TTC^* \\ 0, & otherwise \end{cases} \tag{4}$$

$$TET^* = \sum_{t=1}^{T} TET(t) \tag{5}$$

Where,

Δt: time step length;
TTC^*: TTC threshold value;
$TTC_{(it)}$: the TTC for the i_{th} vehicle at time t;

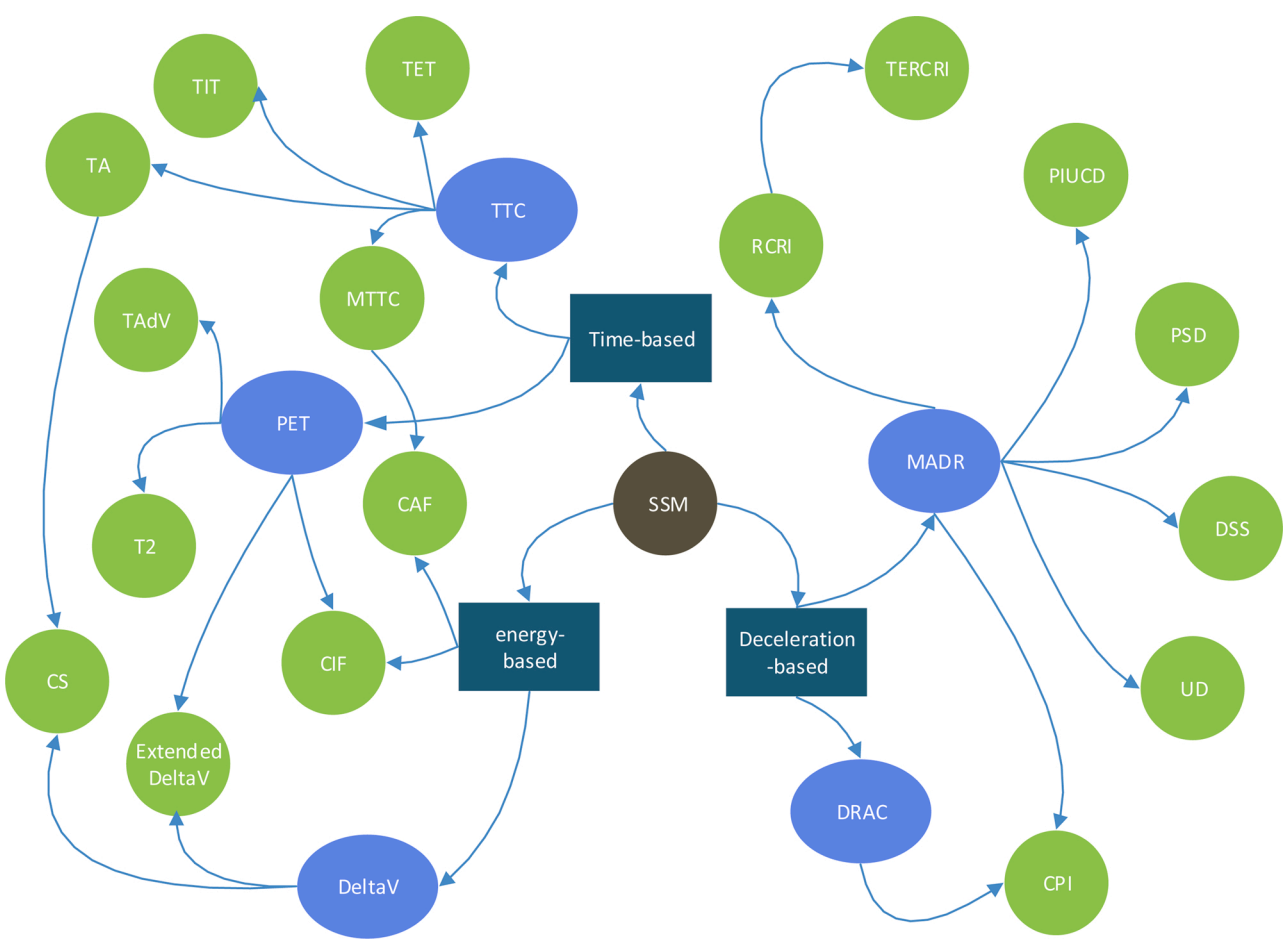

Fig. 1. SSM knowledge map.

n: the vehicle ID;

N: the total number of vehicles; and

δ: switching variable.

Other SSM developed based on TTC include time headway (Vogel, 2003), time to zebra (TTZ; Várhelyi, 1998), time-to-lane crossing (TTL), reciprocal of TTC (i.e., 1/TTC by Chin et al., 1992), Modified Time-to-collision (MTTC) (Ozbay et al., 2008), and T2. Time headway is the elapsed time between the front of the lead vehicle passing a point on the roadway and the front of the following vehicle passing the same point. TTZ indicates the TTC between a crossing pedestrian and a vehicle that is approaching the crosswalk. TTL is defined as the time duration available for a driver before any lane boundary crossing. MTTC predicts TTC of a car-following conflict by considering the accelerations of both the lead and following vehicles, instead of assuming constant velocity. T2 is the predicted arrival time of the second road user (e.g., vehicle, bicyclist), calculated while the first road user has not left the conflict point yet (Laureshyn et al., 2010). When both road users are on a collision course, T2 is equal to TTC.

Post-Encroachment Time (PET) is another important time-based SSM, which refers to the time lapse from the moment that the first vehicle departs a conflict point to the moment that the second vehicle approaches that point. PET was first proposed by Allen et al. (1978), which has no speed and direction assumptions and requires no assumption of the collision course like TTC.

$$PET = t_2 - t_1 \tag{6}$$

Where,

t_2: arriving time at a conflict point of the 2nd vehicle; and

t_1: time of the 1st vehicle departing the conflict point.

Based on the PET definition, additional safety indicators have been proposed. For example, gap time (GT) was proposed to measure the time difference between the entries into the conflict point of two vehicles. The time advantage (TAdv) can be considered as an extension of the PET concept as well (Laureshyn et al., 2010), which is the predicted PET value assuming that the two road users continue with their initial paths and speeds.

The above-mentioned time-based SSM are summarized in Table 2. In general, time-based SSM measure the time proximity to a crash occurrence. Certain thresholds are required for them to identify "crash" conditions or high-risk interactions.

2.2.1.2. Deceleration-based SSM. Instead of measuring time proximity, deceleration-based SSM focus on how vehicles' deceleration can prevent crash occurrence. As early as in 1976, Cooper and Ferguson (1976) proposed DRAC (Deceleration Rate to Avoid the Crash) to determine the severity of an interaction. DRAC is the minimum braking rate required for a vehicle to avoid collision with another vehicle. The calculation of DRAC (see Eq. (7) below) is based on the assumption that one vehicle takes evasive actions while the other retains its speed and direction. To determine the risk of collision, some thresholds are also needed for DRAC.

$$DRAC_{i,t+1} = \frac{(V_{i,t} - V_{i-1,t})^2}{2\left[\left(P_{i-1,t} - P_{i,t}\right) - L_{i-1}\right]} \tag{7}$$

Where,

t: time interval;

P: position of a vehicle (i = the following vehicle, $i-1$ = the lead vehicle);

L: vehicle length; and

V: velocity.

Cunto (2008) further extended DRAC and developed a Crash Potential Index (CPI) by considering a vehicle's braking capability or maximum deceleration rate (MADR). CPI represents the probability that DRAC exceeds the maximum deceleration rate (MADR) at a moment.

Table 2
Summary of major time-based SSM.

Indicator	Definition	Limitations	Advantages
Time to collision (TTC)	The time until a collision between the vehicles would occur if they continued on their current speed and direction.	1. Constant velocity assumption (no evasive actions considered); 2. Require a threshold to determine interaction severity (Threshold sensitive)	1. Frequently used; 2. Easy to measure.
Time to Accident (TA)	Time that remains to a crash when one of the road users starts an evasive action by maintaining the current speed and direction.	1. Relying on the observation of evasive actions; 2. Threshold sensitive	Easy to measure.
Time Exposed Time-to-Collision (TET)	Summation of all moments that a driver approaches a front vehicle with a TTC below a threshold value	1. No risk variation among different values; 2. Threshold sensitive; 3. Require continuous calculation.	1. Suitable for microscopic simulation; 2. Provide an aggregated risk over time.
Time Integrated Time-to-Collision (TIT)	Integral of the TTC profile during the time it is below a certain TTC threshold	1. Difficult to interpret its meaning; 2. Threshold sensitive.	1. Suitable for microscopic simulation; 2. Provide aggregated risk over time; 3. Provide risk variation for different values.
Post-Encroachment Time (PET)	The time lapse between the moment that a road user leaves the conflict area and the other road user arrives the area	1. Applicable for crossing/angle interactions; 2. Cannot reflect changes with the dynamics of interactions; 3. Threshold sensitive.	1. Reflective of driver behaviors; 2. Easy to measure.
Headway (H)	The elapsed time between the front of the lead vehicle passing a point on the roadway and the front of the following vehicle passing the same point	1. Related to following case; 2. Do not account for lateral movement caused by overtaking and lane changing.	Easy to measure.
Gap time (GP)	The time between the entries into the conflict spot of two vehicles	1. Applicable for crossing/angle interactions.	Easy to measure
TAdv	The predicted PET value, provided that the road users continue with their paths and speeds	1. Applicable for crossing/angle interaction; 2. Depending on the assumption of constant speed and direction; 3. Threshold sensitive.	Can be continuously measured as a risk indicator.
Modified Time-to-collision (MTTC)	Modified models which considered all of the potential longitudinal conflict scenarios due to acceleration	1. Difficult to continuously collect data of vehicles' instant accelerations; 2. Not fit for lane changing or head-	1. More advance than TTC; 2. Consider driving discrepancies;

(continued on next page)

Table 2 (*continued*)

Indicator	Definition	Limitations	Advantages
	or deceleration discrepancies	on collision; 3. Threshold sensitive.	

MADR depends on vehicle type and environment conditions such as pavement skid resistance. CPI can be calculated as:

$$CPI = \frac{\sum_{t=t_i}^{tf_i} P(DRAC_{i,t} \geq MADR)\Delta t}{T_i} \tag{8}$$

Where,

MADR is the maximum braking rate;

Δt is time step length;

T_i is the total travel time; and

t_i and tf_i are the initial and final time steps, respectively.

Some studies differentiated distance-based SSM from deceleration-based SSM. In essence, distance-based SSM can also be considered as deceleration-based SSM, since they are both based on assumptions of MADR. The rear-end collision risk index (RCRI) was proposed to identify dangerous conditions by comparing the stopping distance of the lead and following vehicles (Oh et al., 2006), assuming the lead vehicle takes an emergency stopping maneuver with the maximum deceleration rate (i.e., MADR). RCRI can be written as:

$$SSD_L = S + \frac{v_L^2}{2d_m} \tag{9}$$

$$SSD_F = v_F t_d + \frac{v_F^2}{2d_m} \tag{10}$$

$$RCRI = \begin{cases} 0(safe), \ if\ SSD \\ 1(dangerous), \ other\ wise \end{cases} \tag{11}$$

Where SSD_L and SSD_F are the stopping distances of the lead and following vehicles, respectively, v_F and v_l are the speeds of the flowing and lead vehicles, respectively; t_d is the time delay; S is the clearance distance; and d_m is the maximum deceleration rate.

Based on RCRI, the time exposed rear-end crash risk index (TERCRI) was proposed to measure the aggregated risk over time (Rahman and Aty, 2018):

$$TERCRI(t) = \sum_{n=1}^{N} RCRI_n(t) \times \Delta t \tag{12}$$

$$TERCRI = \sum_{t=1}^{Time} TERCRI(t) \tag{13}$$

Where Δt is the time step length. Other distance-based SSM include: Potential Index for Collision with Urgent Deceleration (PICUD) (Uno et al., 2002), Proportion of Stopping Distance (PSD) (Astarita et al., 2012), Difference of Space distance and Stopping distance (DSS) (Okamura et al., 2011), Unsafe density (UD) (Barcelo et al., 2003). They all rely on the assumption of emergency deceleration rates. Among them, PIUCD is calculated as:

$$PIUCD = \frac{v_l^2 - v_f^2}{2a} + S_0 - v_f \Delta t \tag{14}$$

Where,

v_l and v_f: velocity of lead and following vehicles, respectively

S_0: distance between the lead and following vehicles;

Δt: the following drivers' reaction time;

a: emergency deceleration rate to stop.

PSD measures the ratio between the remaining distance to the potential point of collision and the minimum acceptable stopping distance. It is formulated as:

$$PSD = \frac{D_{i,t}}{v_{i,t}^2/2MADR} \tag{15}$$

Where $D_{i,t}$ is the available distance between vehicles.

DSS was defined by the difference of the space and stopping distance, which can be derived by:

$$DSS = \left(\frac{v_l^2}{2\mu g} + d_2\right) - (v_f^2 \Delta t + \frac{v_f^2}{2\mu g}) \tag{16}$$

Where μ is the friction coefficient; g is the gravity acceleration; and d_2 is the distance between the two vehicles.

UD was developed to consider the severity of a potential rear-end crash if the leading vehicle decelerates with maximum braking capacity.

$$UD = (v_l - v_f) v_f R_d \tag{17}$$

$$R_d = \begin{cases} \frac{b}{b_{max}} \ b < 0 \\ 0 \ b \geq 0 \end{cases} \tag{18}$$

Where b is the deceleration rate of the lead vehicle, b_{max} is the maximum possible deceleration rate of the lead vehicle, and v_l and v_f are the speeds of the lead and following vehicles, respectively. Other notable deceleration-based SSM include Deceleration Rate (Malkhamah et al., 2005) and Deceleration-to-Safety Time (Topp, 1998) (Table 3).

2.2.1.3. Energy-based SSM. Unlike time- and deceleration-based SSM which measure the crash proximity of a conflict, energy-based SSM instead were proposed to measure the severity of an interaction. Among them, DeltaV needs to be paid special attention to. It measures the change in velocity forced on road users because of a collision. It depends on the speed and the mass of each road user involved and the angle at which the road users approach each other (Shelby, 2011). Based on momentum conversation assumption (i.e., an inelastic crash), DeltaV can be calculated as:

$$\Delta v_1 = \frac{m_2}{m_1 + m_2}(v_2 - v_1) \tag{19}$$

$$\Delta v_2 = \frac{m_1}{m_1 + m_2}(v_1 - v_2) \tag{20}$$

Where v_1 and $v_2$are the pre-crash velocities for the two involved vehicles with potential collision course; and m_1 and m_2 are the masses of the two vehicles.

Based on DeltaV, some other energy-based SSM have been developed. Bagdadi (2013) proposed conflict severity (CS), which is an integrated indicator combining DeltaV, TA and an assumed maximum average deceleration. The TA and the maximum average deceleration are used to estimate the effectiveness of the evasive actions taken by the involved road users. Another indicator based on DeltaV is the extended DeltaV indicator proposed by Laureshyn et al. (2017a,b). It combines DeltaV with a time indicator and a deceleration constant to estimate the collision probability as well as the potential severity.

Two other notable energy-based SSM are crash index (CAI) (Ozbay et al., 2008) and conflict index (CFI) Alhajyaseen (2015). Instead of calculating DeltaV, they incorporate some other kinetic energy terms. More specifically, CAI estimates the kinetic energy involved in a car-following interaction based on acceleration, speed and MTTC:

$$CAI = \frac{(V_f + a_f MTTC)^2 - (V_L + a_L MTTC)^2}{2} * \frac{1}{MTTC} \tag{21}$$

Where V_L, V_f, a_L, and a_f are the speeds and accelerations of the lead and following vehicles, respectively.

Table 3
Summary of Major deceleration-based SSM.

Indicator	Definition	Limitations	Advantages
Deceleration Rate to Avoid the Crash (DRAC)	Differential speed between a following vehicle and its corresponding lead vehicle divided by their closing time	1. Only applicable for longitudinal car following cases; 2. Need certain thresholds to determined severity of interactions (Threshold sensitive).	Easy to measure
Rear-end collision risk index (RCRI)	Rear-end collision risk index	1. Only applicable for longitudinal car following cases; 2. Rely on a certain boundary to determine interaction severity (i.e. MADR threshold)	Risk Can be continuously measured
Time exposed rear-end crash risk index (TERCRI)	Rear-end collision risk index	Only applicable for longitudinal car following cases	Measure the aggregated risk over time
Crash Potential Index (CPI)	Probability that a given vehicle DRAC exceeds its maximum available deceleration rate (MADR) during a given time interval	1. Only applicable for longitudinal car following cases; 2. Require continuous observation; 3. Rely on a certain boundary to determine interaction severity (i.e. MADR threshold)	1. Suitable for simulation; 2. Provide risk over time; 3. Provide variations in risk for different CPI values
Potential Index for Collision with Urgent Deceleration (PICUD)	Distance between the two vehicles considered when they completely stop with emergency braking	1. Mainly applicable in lane change condition; 2. threshold values yet to be sated up; 3. Do not consider lateral interactions.	Suitable for evaluating risk of crash of the consecutive vehicles.
Proportion of Stopping Distance (PSD)	Ratio between the remaining distance to the potential point of collision and the minimum acceptable stopping distance.	1. Based on evasive actions.	1. Can evaluate single conflict; 2. Easy to measure
Difference of Space distance and Stopping distance (DSS)	DSS is defined by the difference of the space and stopping distance.	Provide information on the number of unsafe vehicle but cannot consider the degree of danger as well as the duration.	1. Can evaluate single conflict; 2. Easy to measure.
Unsafe Density (UD)	Level of "unsafe" in the relation between two consecutive vehicles on the road for a determined simulation step	1. Hard to interpret the meaning, mainly for comparison; 2. Only for longitudinal car-following interaction.	Proposed for microscopic simulation

CFI combines PET with the speeds, masses and angles of the involved road users to estimate the released kinetic energy in a collision. It is designed to consider both crash probability and consequence (i.e., severity).

$$CFI = \frac{\alpha \Delta K_e}{e^{\beta PET}} \tag{3}$$

Where α represents the percentage of the released energy that will affect vehicle occupant(s), ΔK_e is the change in total kinetic energy before and after the crash, and $e^{\beta PET}$ is used to weight conflicts depending on the probability of a crash to occur.

2.2.2. SSM-based models

SSM discussed in Section 2.2.1 use pre-determined thresholds to identify traffic conflicts (which are statistically connected to crashes) from interactions among road users. This approach is known to be subjective and threshold-sensitive. On the other hand, SSM-based models attempt to directly link each traffic conflict to either crash or non-crash, by estimating its crash risk/probability. To our best knowledge, there are two kinds of SSM-based models: uncertainty models and extreme value models.

The idea of uncertainty models was initially introduced by Davis et al. (2011). They showed that in addition to motion estimation, the variance in drivers and vehicles are also important factors that should be considered in predicting crashes. For the same interaction, different combinations of people/vehicles could lead to distinct results. By considering this uncertainty in drivers and vehicles and measuring the general trend (e.g., average), the safety implication of an interaction can be more accurately modeled.

The general uncertainty modeling framework can be described as:

$$P(crash) = 1 - \bigcup_{1}^{N} P(A_i) \tag{4}$$

Where A_i represents the i_{th} necessary condition for crash avoidance (e.g., braking rate, reaction time, steering rate). During an interaction, if all necessary conditions are satisfied, crash will be avoided. Otherwise, crash may occur with probability $P(crash)$. When the probabilities of those necessary conditions are independent, the framework can be rewritten as:

$$P(crash) = 1 - \prod_{1}^{N} P(A_i) \tag{5}$$

To calculate the probability of a necessary condition being satisfied, the distribution of the population needs to be identified. The probability can then be either calculated by integration (i.e., analytical way) or approximated through stochastic models (e.g., random sampling).

Based on this general framework, several SSM-based models have been developed. Saunier and Sayed (2008) proposed a crash probability model based on motion prediction methods. Wang and Stamatiadis (2014a) introduced a Monte-Carlo stochastic process to quantify the crash probabilities of simulated conflicts, considering the variances in drivers' reaction abilities and vehicles' braking capabilities. The Monte-Carlo process can be applied to three main conflict types (crossing, rear-end and lane-changing), covering most real-world conflict cases (Wang and Stamatiadis, 2014b). Wang and Stamatiadis (2016) further conducted sensitivity analyses to explore how drivers' reaction time distribution may affect SSM and whether there is room for improvement in crash predictions. Kuang et al. (2015) also proposed a tree-structured model for estimating rear-end crash probability based on the general uncertainty modeling framework.

Recently, Extreme Value Theory (EVT) based models have also been used to estimate the crash probability of vehicle interactions, which relies on the assumption of extreme value distributions. There are two approaches for applying EVT: block maxima and peak over

(Songchitruksa and Tarko, 2006). For the block maxima approach, observations are aggregated into fixed blocks over time and the maximum of each block is considered as an extreme. Those extremes follow the generalized extreme value distribution shown below:

$$F(x) = \begin{cases} \exp\left\{ -\left[1+\varepsilon\left(\frac{x-\mu}{\sigma}\right)\right]^{-\frac{1}{\varepsilon}} \right\}, \varepsilon \neq 0 \\ \exp\left[-\exp\left\{ -\left(\frac{x-\mu}{\sigma}\right) \right\} \right], \ \varepsilon = 0 \end{cases} \tag{25}$$

Where μ is the location parameter, σ is the scale parameter, and ε is the shape parameter.

For the peak over approach, a threshold u is determined and the threshold exceedance is calculated as $y = x - u$. y follows the generalized Pareto distribution as:

$$F(y) = \begin{cases} 1-\left(1+\varepsilon\frac{y}{\sigma}\right)^{-\frac{1}{\varepsilon}}, \varepsilon \neq 0 \\ 1-\exp\left(-\frac{y}{\sigma}\right), \varepsilon = 0 \end{cases} \tag{26}$$

Songchitruska and Tarko (2006) applied the block maxima approach EVT to evaluate signalized intersection safety, by assuming that the observed PET (i.e., variable x) follows the generalized extreme value distribution in Eq. (25). The crash probability (CR in Eq. (27) below) for each observation can be estimated when PET is equal to or less than 0 s (i.e., a crash):

$$\mathrm{CR} = \Pr(Z \geq 0) = 1 - F(0) \tag{27}$$

Where CR is the risk of crash, Z is the negated PET, F is the generalized extreme value distribution or the generalized Pareto distribution depending on which EVT approach is used. By assuming that the traffic conflict observation period t is representative for a long period T, the estimated crashes N_t can be calculated as:

$$N_t = \frac{T}{t}R \tag{28}$$

Songchitruska and Tarko (2006) reported a high level of consistency between estimated and actual crashes. Since then, a number of univariate EVT safety studies has been conducted considering SSM such as PET (Zheng et al., 2014a,b; Wang et al., 2018) and TTC (Åsljung et al., 2017; Farah and Azevedo, 2017; Tarko, 2018; Orsini et al., 2019, 2020). In particular, Åsljung et al. (2017) analyzed 250,000 km driving data using the EVT framework, and concluded that EVT was a promising tool for CAV safety evaluation. Zheng et al. (2018) and Wang et al. (2019a,b) further extended the univariate EVT framework to bivariate and used it to evaluate the safety of freeway entrances and signalized intersections, respectively. Zheng et al. (2018) modeled the combination of PET and the length proportion of merging, and reported promising results. Wang et al. (2019a,b) modeled the combinations of four SSM: TTC, TA, PET and maximum deceleration. According to their results, the combination of TA and PET appeared to be the best. Since then, bivariate EVT models have been applied in several other safety studies, in which various SSM combinations were examined, including TTC and PET (Zheng and Sayed, 2019), TTC/PET and DRAC (Zheng et al., 2019a), and TTC and distance headway (Cavadas et al., 2020). The uncertainty and extreme value SSM models are summarized in the table below (Table 4).

3. SSM for CAV safety studies

This section focuses on reviewing safety studies using SSM and simulation, and how SSM have been utilized for CAV safety evaluation. We start with analyzing the limitations of field observation-based SSM, followed by discussion on the widely adopted approach of using simulation-based SSM for safety studies. We then summarize how simulation-based SSM have been utilized in evaluating CAV safety (Table 5).

Table 4
Summary of energy-based SSM.

Indicator	Definition	Limitations	Advantages
DeltaV	The change in velocity between pre-collision and post-collision trajectories of a vehicle	1. Do not consider evasive actions (no speed/direction change from pre- to post crash) 2. Inelastic collision assumption.	Evaluate the consequence of interactions.
Extended DeltaV	An extension of DeltaV incorporating crash proximity	Constant deceleration assumption	Estimate both the proximity and consequence of interactions.
Conflict severity (CS)	An extension of DeltaV by incorporating TA and maximum average deceleration	Maximum deceleration assumption	Estimate both the proximity and consequence of interactions.
Conflict index (CFI)	It combines PET with the speed, mass and angle of the involved road users to estimate the released kinetic energy in a collision	1. Require parameter calibration; 2. Hard to interpret meaning;	Estimate both proximity and consequence of interactions.
Crash index (CAI)	Influence of speed on kinetic energy involved in a potential collision	1. Only for rear-end; 2. Requirement of the data collection on vehicles' accelerations	Estimate both proximity and consequence of interactions.

3.1. Field observation-based SSM

SSM were initially developed based on field safety studies (i.e., conflict studies). Field observations are often time-consuming and labor intensive. Moreover, human observation errors could be introduced and affect the reliability of field-based safety studies. To address these issues, computer vision and various sensor techniques (Ismail et al., 2010; Autey et al., 2012; Laureshyn et al., 2017a,b; Wu et al., 2018; Fu et al., 2016; Machiani and Abbs, 2016; Chen et al., 2017) have been introduced to continuously detect and track vehicle motions, without much human intervention. Ismail et al. (2010) conducted a conflict-based before-after (BA) studies on the safety impact of a scramble phase

Table 5
Summary of SSM-based models.

Model	Description	Cons	Pros
Uncertainty model	Measure the proximity of crash based on a random stochastic model, incorporating driver and vehicle uncertainty	1. Could be biased due to inaccurate description of uncertain factors.	1. Proposed for multiple interaction types; 2. Consider uncertainty of evasive actions; 3. Generative model estimations
Extreme Value Theory model	Model the extreme case of interactions based on extreme distributions	1. Require assumptions on extreme distributions; 2. Require thresholds either for block maxima or peak over methods; 3. Require enough samples to reduce estimation bias	1. Can be applied on multiple indicators; 2. Link the interaction to crash statistically

treatment, by an automated video technique. Four SSM were used: TTC, PET, DST, and GT. Based on the similar approach, Autey et al. (2012) evaluated the safety implication of right-turn smart channels. Laureshyn et al. (2017a,b) applied three approaches to derive pedestrian SSM using intersection videos. The three approaches were the Swedish traffic conflict technique (Swedish TCT), the Dutch conflict technique (DOCTOR) and the Canadian probabilistic surrogate measures of safety (PSMS) technique. TCT and DOCTOR manually count critical traffic conflicts based on time-based SSM (e.g., TTC, PET), while PSMS considers probabilities of vehicle trajectories to estimate potential crashes. PSMS relies on video processing techniques to automatically track road users. Overall, the three methods generate similar results. However, the PSMS method needs to be further improved in deriving cyclists related SSM due to inaccurate trajectories extracted. To improve SSM extraction in low light and adverse weather conditions, Wu et al. (2018) developed a LiDAR-based approach to obtain trajectories of all road users at intersections. They focused on vehicle-pedestrian near-crash identification and proposed two SSM: Time Difference to the Point of Intersection (TDPI) and Distance between Stop Position and Pedestrian (DSPP). Fu et al. (2016) used videos from a thermal camera to derive SSM for pedestrians at unsignalized crosswalks and found it to work well under low visibility conditions. SSM and safety indicators considered include vehicle approaching speed, post-encroachment time (PET), yielding compliances, conflict rates, and pedestrian exposure. Machiani and Abbas (2016) developed a TTC-based histogram to evaluate the safety of dilemma zone based on radar data. Chen et al. (2017) utilized drone collected videos at an intersection and derived Post-encroachment Time (PET) and Relative Time to Collision (RTTC) to analyze vehicle-pedestrian collision risk.

Leveraging the computer vision and sensor technologies, human labor requirements can be significantly reduced and data accuracy can be improved for field-based safety studies. However, those techniques are relatively complicated and require well-trained safety analysts to apply. Moreover, observation errors can still happen under certain conditions, such as occlusion by large vehicles, adverse weather and poor visibility, and poor lighting conditions. Recent advancements in deep learning have substantially improved the accuracy of computer vision-based object detection and tracking. Some commercial software products have already been developed for video-based safety evaluation. Although a detailed analysis and comparison of these deep learning-based computer vision algorithms/products is out of the scope of this paper, such algorithms/products demonstrate great potential to address the aforementioned concerns about computer vision and sensor technologies, making field observation-based safety evaluation much more viable and SSM more important than before. Compared to SSM generated based on traffic simulations (see Section 3.2 below), SSM from field observations are more realistic. As the penetration rate of CAV increases, field observation can be an important approach to derive SSM for evaluating CAV safety.

3.2. Simulation-based safety studies using SSM

Simulation tools have been extensively utilized in traffic analysis. Initially, traffic simulation was mainly for operational evaluations. Compared to field-based studies, simulation tools are able to build traffic scenarios quicker and easier and allow for comparison of different strategies under the same traffic input. Due to the advantages offered by simulation, many researchers also attempted to conduct safety evaluations using SSM based on traffic simulations.

Despite the advantages, there are doubts about simulation-based safety evaluations. Some researchers (Tarko, 2018) criticized that simulation tools are unable to replicate extreme and dangerous vehicle interactions/traffic conditions, since driver behavior models in simulation tools are developed to represent normal driving habits instead of distracted and aberrant behaviors. In this sense, they claimed that simulation-based SSM only reflect traffic exposure instead of true safety impacts. Another issue with simulation-based SSM is that simulated vehicles follow certain preprogrammed paths. For example, vehicles in VISSIM follow links and connectors. If the connectors for two opposing left-turns at an intersection are coded without any overlap, the chance for those left-turn vehicles to have head-on crashes does not exist. In reality, left-turn vehicles may not always follow lane marking and can make wide/narrow turns. Such uncertainty in vehicle turning radius is not adequately considered in existing simulation tools and can generate biased SSM results.

On the other hand, others believed that the theoretical foundation for simulation-based safety studies is valid. Although some extreme cases may not be captured by simulations, a large number of vehicle interactions can still be observed by a well-calibrated simulation model. Studies have shown that by properly calibrating simulation models, the distribution of simulated SSM can be highly consistent with that of field observed SSM (Gettman and Head, 2003; Huang et al., 2013; Ozbay et al., 2008; Zheng et al., 2019b), supporting that SSM based on traffic simulation results are reliable and valid. For instance, Zheng et al. (2019b) applied EVT models on simulated conflicts and reported promising results. Another interesting way of evaluating safety based on simulation is to directly simulate driver errors and identify the resulting crash consequence (Astarita and Giofré, 2019). Such a method also requires proper model calibrations.

3.3. Simulation and SSM for evaluating CAV safety

Since CAV have not been deployed at large scales in the real world, it is difficult to collect field data to explore CAV's safety implications. Currently, microscopic traffic simulation has been the main tool for CAV safety studies. CAV can eliminate driver errors (e.g., distracted driving) but are still affected by potential mechanical and communication errors, software bugs, and sensor malfunctions. Such errors, compared to mistakes made by human drivers, are relatively easier to model by traffic simulations. In this sense, simulating 100 % CAV could produce more reliable outputs than Human-Driven Vehicles (HDV). However, when dealing with a mixed environment with both CAV and HDV, careful simulation model calibration is important for generating reliable safety results.

When applying microscopic traffic simulation tools for CAV safety modeling, specialized software packages such as VISSIM, Paramics, SUMO are often utilized to generate detailed vehicle trajectories. Either the Surrogate Safety Assessment Model (SSAM) or other custom-developed tools are then used to analyze trajectories and compute SSM (FHWA, 2003, 2008). Other than specialized traffic simulation packages, general-purpose simulation tools (e.g., Matlab) have also been adopted for safety analysis, in which the traffic environment and scenarios are much simplified compared to those in specialized microscopic simulation packages. The current literatures on simulation-based safety modeling and SSM applications can be categorized into two groups, (1) safety effects evaluation; and (2) trajectory optimization, as summarized in Tables 6 and 7 respectively.

As for safety effects evaluation, mainly time-based and deceleration-based SSM have been used, and the most popular SSM is TTC (Rahman et al., 2018, 2019a,b; Tibljaš et al., 2018; Li et al., 2018; Virdi et al., 2019; Morando et al., 2018; Papadoulis et al., 2019). For instance, Morando et al. (2018) found that increasing automated vehicles penetration rate could significantly improve traffic safety at both signalized intersections and roundabouts. Other than TTC, TIT and TET have also been frequently used (Li et al., 2017a,b,c, ; Rahman and Aty, 2018; Rahman et al., 2019a,b). Additionally, TA has been adopted (Wu et al., 2019). Applications of deceleration-based SSM include the distributions of hard braking (Zhong et al., 2019), RCRI (Li et al., 2018; Rahman and Aty, 2018; Rahman et al., 2019a,b), sideswipe crash risk (i.e., the number of lane-changing conflicts) (Rahman and Aty, 2018), and TERCRI (Rahman and Aty, 2018). Other safety indicators have also been applied for CAV safety effects evaluation, such as standard deviation of

Table 6
Summary of SSM and other safety indicators used for CAV safety evaluation.

Indicator	Reference	Simulation Tool	Threshold
TTC	Tibljaš et al., 2018	VISSIM + SSAM	1.5 s
	Rahman	VISSIM	1.5 s
	Virdi et al., 2019	VISSIM + SSAM	1.5 s
	Papadoulis et al., 2019	VISSIM + SSAM	1.5 s
	Li et al., 2018	General-Purpose (GP) simulation tools	2 s
	Morando et al., 2018	VISSIM + SSAM	1.5 s
TET, TIT	Jeong et al., 2017	General-Purpose (GP) simulation tools	2 s (TTC)
	Rahman and Aty 2018, Rahman et al., 2019a,b	VISSIM + SSAM	1−3 s (TTC)
	Li et al., 2017a,b,c	VISSIM	2 s (TTC)
RCRI	Rahman and Aty 2018, Rahman et al., 2019a,b	VISSIM + SSAM	3.42 m/s for PC 2.42 m/s for HGV
	Li et al., 2018	General-Purpose (GP) simulation tools	3.4
TA	Wu et al., 2019	VISSIM + SSAM	2 s
DARC	Zhong et al., 2019	VISSIM	−3 m/s^2
Maximum speed	Tibljaš et al., 2018	VISSIM + SSAM	–
Standard deviation of speed	Fu et al., 2019	PreScan	–
	Rahman and Aty, 2018	VISSIM + SSAM	–

speed (Rahman and Aty, 2018; Fu et al., 2019) and Maximum speed (MaxS) (Tibljaš et al., 2018). Note that in most literature, CAV and HDV were evaluated based on the same SSM (e.g., TTC = 1.5 s or 2 s), and no separate SSM for CAV were found in those studies.

In trajectory optimization studies, CAV maneuver decisions (e.g., merging, crossing) and/or trajectories are planned ahead of time and are optimized considering safety constraints consisting of SSM. For maneuver decisions, distance and time gap constraints are often adopted to optimize merging and crossing maneuver safety and to ensure enough safety space between vehicles (Lee and Park, 2012; Xu et al., 2018, 2019, 2020; Dong et al., 2020; Jing et al., 2019; Ren et al., 2020). For trajectory planning, minimum safety headway/gap and TTC constraints have been used to ensure longitudinal safety (Zhao et al., 2017; Duan and Zhao, 2017; Huang et al., 2018; Yao and Friedrich, 2019; Ren et al., 2020). In some studies, SSM have also been used in the CAV longitudinal trajectory optimization objectives (not just as constraints). The adopted SSM and safety indicators include minimum safety time gap/headway (Alonso and Pérez-Oria, 2010; Fernandes and Nunes, 2015; Milanes and Shladover, 2016; Xu et al., 2018; Zhou et al., 2020; Qu et al., 2020), minimum deceleration (Cherian and Sathiyan, 2012), platoon string stability (Roncoli et al., 2015; Flores and Milanés, 2018; Wang, 2018, Wang, 2019; Huang et al., 2019), TET and TIT (Jeong et al., 2017; Rahman and Aty, 2018; Liu et al., 2020), and inverse TTC (Jeong et al., 2017; Qu et al., 2020). Among these studies, string stability was often used as the control objective for CAV platoons with ACC/CACC functions. Although string stability strictly speaking is not considered as SSM, previous literature has proved that better string stability could bring important safety benefits.

4. Discussion

4.1. Developing simulation models for CAV

Microscopic traffic simulation models were introduced initially for modeling vehicle movements following normal traffic rules. The traffic flow models behind various simulation tools are designed to generate typical traffic interactions that frequently occur. However, the essence of SSM lies in a model's ability to also produce infrequent and risky

Table 7
Summary of SSM used for CAV trajectory optimization.

Indicator	Reference	Usage	Simulation	Optimization
Safety distance	Alonso and Pérez-Oria, 2010	Included in objectives as baseline	General-Purpose (GP) simulation tools	Genetic optimization
	Fernandes and Nunes, 2015, 2015	Extra safety distance was used for safety improvement		Frequency Domain Analysis with constant headway policy
	Xu et al., 2018	Included in objectives as soft constraint for performance		quadratic programming
	Lee and Park, 2012; Duan and Zhao, 2017; Huang et al., 2018; Torres and Malikopoulos, 2017, Zhao et al., 2017	Used as constraints		–
	Zhou et al., 2020	Included as Part of Objectives for reward selection		reinforcement learning
Safety headway (gap)	Xu et al., 2019, Xu et al., 2020; Jing et al., 2019; Yao and Friedrich, 2019; Dong et al., 2020, Ren et al., 2020	Used as constraints		–
Traffic flow/ string stability	Flores and Milanes, 2018	String stability for CACC	GP simulation tools	Frequency Domain Analysis
	Wang, 2018	sufficient condition for String stability		Frequency Domain Analysis
	Huang et al., 2019	A penalty term was added to restrict speed in congestion for safety		mean field game
	Roncoli et al., 2015	String stability was used as objectives	AIMSUN	Model predictive control
	Wang, 2019	String stability for CACC controller		Frequency Domain Analysis
	Milanes and Shladover, 2016	String stability for CACC controller	GP simulation tools	Frequency Domain Analysis
	Mohtavipour and Mollajafari, 2019	String stability for CACC controller		Heuristic reference model
Monetized cost	Liu et al., 2020	Safety was converted to monetized cost as a part of objectives		monetized cost
Risk map	Park and Oh, 2019	A new SSM was developed for optimizing	VISSIM	Heuristic method

(continued on next page)

Table 7 (*continued*)

Indicator	Reference	Usage	Simulation	Optimization
		safety with speed control		
TTC	Ren et al., 2020, Zhao et al., 2017	Used as constraints	GP simulation tools	–

interactions. Some researchers (Gettman and Head, 2003; Huang et al., 2013; Ozbay et al., 2008) argued that simulation tools are able to generate valid data for computing SSM through proper model calibration. The calibration process will make simulated road users more (or less) aggressive, and generate simulated interactions that are overall consistent with field observations. Thus, a proper simulation calibration procedure is vital for simulation-based safety studies. Additionally, some human factors (e.g., distraction, non-compliance of traffic rules) are not well defined in many simulation tools. With better knowledge of driver behaviors that contribute to crashes and near crashes, these human factors could be properly modeled and incorporated into simulation tools, which will make simulation-based safety outputs more realistic and reliable after careful calibration.

Regarding CAV related safety research, simulation by far is the most viable choice for many researchers without access to CAV. For fully automated vehicles without human in the loop, theoretically the behaviors and mechanism of CAV can be well captured by simulation, since there is no need to consider human factors such as reaction time and distracted driving. However, research is still needed to calibrate simulation models for modeling mixed autonomy traffic to accurately reflect CAV behaviors and human driver reactions to CAV. In this case, traditional and virtual reality (Xie et al., 2018) driving simulators could play an important role by bringing human drivers into the loop and model how CAV and human drivers interact with each other in a simulated environment. Another option is to use data from pilot connected vehicles deployment projects and automated vehicles field tests, which is detailed later in Section 4.7.

As for simulation tools, general-purpose simulation software (e.g., Matlab) has been often used for safety optimization while specialized traffic simulation software packages (e.g., VISSIM) have been mainly used for safety impacts evaluation. Safety studies based on specialized traffic simulation tools require considerable efforts for model development and calibration, while they could generate more detailed and accurate results than general-purpose simulation tools, and are more suitable for safety impacts evaluation. On the other hand, general-purpose simulation tools are computationally less demanding and can be well integrated with complex optimization algorithms to identify promising safety strategies for further detailed analyses using specialized simulation models. However, simulations using general-purpose tools often make simplified assumptions and may ignore important aspects of vehicle/traffic characteristics and vehicle interactions. Therefore, the choice of simulation tools is essentially a trade-off between accuracy and efficiency. It is also an interesting research topic that deserves further investigation so that guidelines can be developed for selecting the most appropriate modeling tools and/or for developing new tools for simulation-based CAV safety studies. For example, a hybrid simulation tool might be developed that considers simplified traffic models for roadway segments (focusing on rear-end and sideswipe crash risk) while more detailed models for intersections, ramps, and driveways.

4.2. The validity/transferability of SSM

A critical issue regarding SSM is its validity. In previous literature, substantial efforts have been devoted to validating SSM. However, those studies were performed considering traditional traffic environments with only Human Driven Vehicles (HDV). When modeling safety in mixed autonomy traffic or fully automated traffic, it is unclear whether the SSM validated in traditional traffic environments can still be applicable. On one hand, the behaviors of CAV can be very different from those of HDV. Even among CAV with different automation/connectivity levels, their behaviors are likely to differ. On the other hand, the behaviors of HDV can change in response to the existence of CAV. Given the new situation, conventional SSM or SSM-based models, may need to revise accordingly. For instance, the thresholds used in SSM may need to be adjusted. Specifically, can TTC = 1.5 s still be a good choice for both HDV and CAV? Another valid question is that for SSM-based models, are those underlying models transferable to the new traffic environment? For example, does extreme value distribution calibrated in traditional traffic environment still hold for a 100 % CAV environment or mixed autonomy traffic?

Based on our review, few research (e.g., Weng et al., 2020) so far has attempted to address this issue. Most existing CAV safety studies assume that conventional SSM are still valid and transferable to CAV environment. For future simulation-based CAV safety studies, the validity of SSM needs to be carefully examined and compared with field data when available. In addition, the transferability of SSM deserves attention. It is important to understand whether an SSM can provide reliable results across various traffic environments (e.g., all human vs. mixed autonomy traffic), infrastructure types, traffic compositions, weather conditions, etc.

4.3. SSM and SSM-based models

Traditionally, SSM depend on certain thresholds to identify risky interactions linked to crashes. Generally, they are easier to calculate compared to SSM-based models, but have obvious shortcomings. For example, safety studies typically set TTC threshold to 1.5 s to identify risky interactions. This implies that all interactions with TTC values greater (or less) than 1.5 s are considered equally dangerous (i.e., resulting in the same crash severity level), if TTC is the only SSM considered. TTC does not account for any potential evasive actions that are likely being taken during the course of an interaction. Also, in reality, drivers have varied reaction times and vehicles have different braking performance. Consequently, interactions with the same TTC can be associated with quite different levels of crash risk and severity outcomes depending on driver and vehicle characteristics, type of interaction, evasive actions, etc. Even for the same interaction type, different vehicle speeds with the same TTC can pose various levels of difficulty for drivers to avoid the crash as well as lead to diverse levels of crash severity. Although some TTC variations (e.g., TIT) could estimate the severity of an interaction, a TTC threshold is still required as a precondition. Similarly, the downside of DRAC (or its variations) is that traffic interactions are divided into groups based on a set of threshold values (Cunto, 2008). Those thresholds are important but often require additional evaluations to find their optimal values. For future studies, the thresholds for SSM need to be examined very carefully.

SSM-based models can directly estimate an interaction's crash risk or probability without explicitly setting thresholds, which is an important advantage over conventional SSM. However, bias could still be introduced by SSM-based models. The reviewed SSM-based models may not be able to capture all potential crash risk impacts of an interaction due to either their inherent statistical nature or external factors that are not included. For example, a univariate EVT only captures certain crash risk aspects of an interaction since it measures just one indicator. More specifically, an EVT SSM based on TTC may only be able to measure the risk in terms of time proximity, while other risk aspects (e.g., speed, mass, deceleration, etc.) are not reflected. Therefore, it is worthwhile to explore more advanced solutions such as bivariate EVT models (Wang et al., 2019a,b). Even if this EVT SSM is very comprehensive and can cover all risk aspects, its EVT distribution is still an approximation of the underlying true pattern. The parameters of the fitted EVT distribution could be biased due to partial/inaccurate observations and limited

sample size. All these factors may lead to biases in EVT model estimation and inaccurate SSM-based models, thus should be handled with care.

4.4. Specific SSM for CAV

It can be expected that the mixed autonomy traffic environment will emerge sooner than a 100 % CAV environment. The behaviors of Human-Driven Vehicles (HDV) are different from those of CAV, and can also be affected by CAV. Compared to HDV, CAV have much shorter reaction time (approaching zero depending on CAV system design) and are able to share precise and complex information (e.g., vehicle maneuvers) in real time with each other and with infrastructure. Thus, even if conventional SSM are still applicable for evaluating CAV safety, it is questionable whether they are accurate enough. Therefore, dedicated SSM need to be developed for CAV at different automation/connectivity levels, based on field data or driving simulator results.

For example, vehicle platooning is a promising and viable solution for improving mobility and safety. A well-known vehicle platooning technique is Cooperative Adaptive Cruise Control (CACC), which is enabled by vehicle connectivity and Level 1 automation. In mixed autonomy traffic, a vehicle platoon could include both HDV and CAV at different connectivity and automation levels. As such, the longitudinal safety of each vehicle could differ, depending on its ability of sensing the surrounding and reacting to potential risk. Many CACC studies are focused on examining string stability, which can also be considered as a safety performance indicator for the whole platoon. However, the underlying relationship between string stability and individual vehicle's safety is not explicitly understood and needs to be further explored.

In addition, SSM for specific traffic scenarios may also need to be investigated, such as lateral safety related to lane-changing and merging maneuvers. Measuring lateral safety is important for both HDV and CAV, especially in mixed autonomy traffic. For example, when modeling cooperative lane changing/merging scenarios with CAV (Ren et al., 2020, 2021), a small headway may not necessarily lead to collision if all vehicles involved are fully connected and aware of each other's next move. Based on the literature review, research on SSM for lateral risk caused by lane changing and merging maneuvers is very limited (Vogel, 2003; Wang and Stamatiadis, 2013; Kanagaraj et al., 2015).

4.5. Universal SSM

Since CAV and HDV may coexist for an extended period and given the above issues, it is interesting to research whether a universal set of SSM can satisfy the needs of simulation-based CAV safety studies under mixed traffic environment. In this environment, CAV differ from HDV in terms of behaviors and capabilities. Moreover, CAV can have various automation levels. Even at the same automation level, CAV manufactured by different companies are likely to demonstrate quite different behaviors. Additionally, users may be able to set the driving behavior of a CAV to aggressive mode, cooperative mode, etc. If different SSM are adopted for different road users (e.g., CAV, HDV), it could be difficult for decision making due to two main reasons: (1) How to combine different SSM into an aggregate value for plan/design comparison; and (2) How to weight different SSM. In this case, a universal set of SSM would be helpful to assist decision makers with identifying plans of maximum safety benefits during transportation planning, infrastructure design, traffic control and management, etc.

4.6. SSM for safety-oriented CAV trajectory optimization

CAV can plan their trajectories (e.g., lane changing and car following) ahead of time. When planning such trajectories, safety is always an important consideration. SSM-based models are suitable for such a purpose, since they can be modified to continuously monitor and predict a CAV's crash risk status by considering uncertain aspects such as sensor malfunctions, cyber-attack, communication latencies, signal transmission range limitations, data packet loss, vehicle dynamics, and traffic flow disturbance and other dangerous events (e.g., pedestrian jaywalk).

For the constraints used in trajectory optimization, most literature adopted a simple static safety distance boundary (Zhao et al., 2017; Duan and Zhao, 2017; Yao and Friedrich, 2019; Ren et al., 2020). In other words, a minimum distance has to be maintained at each time step to ensure safety. To improve the safety and robustness of CAV trajectory control, the static safety distance boundary may be made time- and environment-dependent, taking factors such as vehicle speed into consideration. In addition, lateral positions planning can be incorporated into longitudinal trajectory optimization, which is particularly important for lane-changing and merging maneuvers. For instance, SSM related to lateral crash risk can be included in either the objective function or some constraints.

4.7. New data sources for developing SSM

Besides field observation using various sensors (see Section 3.1), it is anticipated that some new data sources such as connected vehicles and even automated vehicles will play an important role in developing SSM. For instance, He et al. (2018) used the data from the Safety Pilot Model Deployment (SPMD) (Gay and Kniss, 2015) to develop SSM. The SPMD study was conducted in Ann Arbor and included about 3000 participating vehicles and 30 roadside equipment (RSE) installed mostly at signalized intersections. Those vehicles broadcast Basic Safety Messages (BSMs) containing vehicle speed, location, etc. at 10 Hz. Based on the SPMD data, they calculated MTTC, TTC, and DRAC and correlated them with historical crash records. Also based on the SPMD data, Xie et al. (2019) proposed a new SSM called Time to Collision with Disturbance (TTCD). Compared to conventional TTC, TTCD can better consider the risk in cases when the following vehicle is slower than but very close to the leading vehicle. Under such circumstances, a small speed disturbance may lead to a crash. Guo et al. (2010) considered near-crash events derived from the SHRP2 Naturalistic Driving Study (NDS) data (Blatt et al., 2015) as a crash surrogate, which is defined as "Any circumstance that requires a rapid, evasive maneuver by the participant vehicle, or any other vehicle, pedestrian, cyclist, or animal, to avoid a crash." (Guo et al., 2010). Ishak et al. (2017) also looked at the NDS data and tried to relate a driver's odd of being involved in distracted driving behaviors to five measures: GPS speed, lateral and longitudinal acceleration, throttle position, and yaw rate. Although they were unable to find statistically significant relationships, the results from a neural network model suggest that the five measures are useful in identifying distracted driving.

Stipancic et al. (2018) used driver smartphone GPS data to derive SSM. Since GPS data only contains information for the subject vehicle, hard braking and accelerating events are used to define SSM and are compared to historical crash data. Similarly, Strauss et al. (2017a,b) used cyclist hard braking events extracted from GPS data to analyze bicycle safety. Boonsiripant (2009) used vehicle speed profile derived from GPS data and developed over ten safety indicators, including speed variation, mean of speed band, acceleration noise, stop frequency per trip per mile, etc. For SSM and safety indicators derived based on GPS or speed profile data, properly specifying the thresholds for hard braking and accelerating events is important, and may benefit from taking the corresponding traffic environment (e.g., freeways vs. local roads) into consideration. Following this direction, SSM may also be derived from Waze, INRIX, and smartphone accelerometer data.

As can be seen, recently there has been a considerable amount of interest in using observed data for deriving SSM. Compared to simulated data, observed data better capture the randomness in traveler behaviors and generate more realistic risk measures. These datasets can be broadly classified into three categories:

- Video/Lidar/radar data: These datasets are typically collected from roadside sensors or drones at fixed locations, and analyzed automatically by computer programs. They can capture vehicles, pedestrians, and cyclists and the interactions among them. Depending on where a sensor is mounted, its view might be blocked by obstacles (e.g., trees or trucks). The SSM derived using such datasets and the safety analysis results are often specific to the location where the data is collected, and may not be directly generalized to other sites. Such data provide useful information in studying how built environment (e.g., intersection geometry and control) affects the behaviors of a wide range of travelers under different traffic volume, weather, and lighting conditions.
- GPS and speed profile data: Such datasets may come from the smartphones of drivers and cyclists, electronic logging devices in commercial vehicles, ridesharing companies, insurance companies, companies selling traffic data, etc. They usually cover a large area and provide good opportunities to investigate how traveler behaviors vary as a result of changing environments, and to identify hot spots. Some datasets may also include the information about the data contributors (e.g., age, gender). A challenge in deriving SSM from such datasets is that limited information (other than roadway geometry and traffic control) regarding the surrounding environment (e.g., distance to front vehicle) is available. Also, the sample sizes across locations could be significantly different, which may lead to biased SSM results. Additionally, SSM developed based on GPS and speed profile data can only account for the risk of rear-end crashes (not other types such as side-swipe) due to the limitation of the data.
- Connected vehicles testing data and Naturalistic Driving Study (NDS) data: Notable connected vehicles datasets include the SPMD data (Gay and Kniss, 2015) and the USDOT Connected Vehicles Pilot Study data (USDOT, 2021). The connected vehicles and NDS data are included in the same category due to their similarities. Both types of datasets provide detailed information about the subject vehicle, including distance to obstacle, speed, longitudinal and lateral accelerations, vehicle steering and braking, etc. These detailed information (e.g., steering, lateral acceleration, and distance to obstacle) sets them apart from the above GPS and speed profile data, and make it possible to calculate SSM such as TTC. In addition, the NDS dataset includes videos capturing roadway and driver's face, which are important for detecting distracted driving and dangerous traffic scenarios (e.g., road debris) and associating them with vehicle kinematics. Such relationships may be generalized and used to analyze the more widely available GPS and speed profile data.

Some automated driving technology development companies such as Baidu, Lyft and Waymo have also made their data available. Such datasets are the most comprehensive and capture all traffic surrounding the automated vehicle and the automated vehicle's movement. Different from the NDS data, automated vehicles datasets also include camera and/or Lidar data for the left, right and rear views, allowing us to analyze the risk of side-swipe and rear-end (the automated vehicle being rear-ended by another vehicle) crashes. They may potentially be used to analyze the collision risk among the surrounding traffic (e.g., a nearby car with a pedestrian). Note that automated vehicles supposedly can predict the movements of the surrounding traffic and take proactive actions to avoid collision. Therefore, the estimated SSM associated with the automated vehicle could be biased, since they reflect the behaviors of an extremely safe "driver". In this case, SSM associated with the surrounding traffic may better reflect the typical traffic safety risk. Although this review has not identified any studies using real-world automated driving data for deriving SSM, this certainly is an interesting area for future research.

4.8. New concepts for evaluating future CAV safety

Since CAV are controlled by algorithms, CAV safety performance depends much on how conservative (i.e., safe) the control algorithms are designed to be. Incorporating proper safety principles into CAV control algorithms and thoroughly evaluating their safety performance become very important. Shalev-Shwartz et al. with Mobileye (2018) proposed a Responsibility-Sensitive Safety (RSS) concept, which is a technology-neutral framework consisting of five safety principles that can be adopted by any CAV manufacturers to design CAV control algorithms to ensure consistency in safety. Under this framework, various SSM can be utilized to evaluate CAV safety. Weng et al. (2020) proposed a Model Predictive Instantaneous Safety Metric (MPrISM) for modeling automated vehicles safety. MPrISM is a high-dimensional model predictive TTC metric that estimates the TTC when other road users behave very aggressively while the automated vehicle tries its best longitudinal and lateral maneuvers (compared to only longitudinal in most previous studies) to avoid crash. Nister et al., 2019 developed a Safety Force Field (SFF) model as the basis of CAV control system. Similar to MPrISM, SFF also considers a high-dimensional (e.g., both longitudinal and lateral) model predictive approach to estimate crash risk, and this approach can be used to develop new SSM. If a CAV's perception system works properly, the SFF can guarantee safety. Winner et al. (2019) discussed the PEGASUS project (Project for the Establishment of Generally Accepted quality criteria, tools and methods as well as Scenarios and Situations). As its names suggest, this project focuses on the standards, procedures, testing scenarios, etc. needed for evaluating CAV safety. Additionally, Fraade-Blanar et al. from RAND (2018) recently released a report on the framework and measures for evaluating CAV safety. Some of these studies are not directly related to developing new SSM (e.g., RSS, RAND report, PEGASUS). However, they can help us understand the challenges facing CAV safety, and how to adopt existing SSM and/or develop new ones to evaluate CAV safety. In particular, the high-dimensional MPrISM and SFF may inspire new SSM for analyzing CAV safety using field data generated by automated driving companies such as Waymo.

5. Conclusions

SSM are important for safety performance evaluation, since crashes are rare events and historical crash data does not capture near-crash events that are also critical for improving site safety. SSM have been researched for years and have been widely used in simulation-based CAV safety studies. This paper focuses on reviewing SSM and their applications in CAV safety studies. The main contributions of this paper are three folds. First, we provide a comprehensive review of significant SSM and categorize them into two major categories, SSM and SSM-based models, based on how they are developed and how they estimate the severity of an interaction. Second, we summarize field and simulation-based safety studies using SSM. We further discuss various SSM applications in simulation-based CAV safety studies, including safety impacts evaluation and safety-oriented trajectory optimization. Third, we identify and discuss a series of issues related to SSM for CAV safety studies and point out some directions for future research. We hope the findings can help researchers and practitioners understand the pros and cons of existing SSM and choose the most appropriate one(s) for their safety studies. Also, the analyses may help researchers avoid duplicative research and inspire innovative ideas for future SSM research.

As road users change (e.g., aging driver population) and new users are introduced (e.g., CAV with different levels of automation), there will be interesting and oftentimes challenging research problems. For example, in mixed autonomy traffic HDV could behave differently than in traditional traffic environment, and conventional SSM may not be valid and cannot be adopted without modification. This raises a number of questions, such as "Is single SSM good enough for CAV of different levels of automation and connectivity?", "Can we find a universal SSM for mixed autonomy traffic?", "What SSM are capable of describing the crash risk for a vehicle platoon and individual vehicles in the platoon?" and "What SSM are better for trajectory optimization to improve

safety?" These are some of the questions that the authors believe the traffic safety research community should address in the near future before CAVs are widely deployed.

Author statement

Chen Wang: Conceptualization; Methodology; Writing - Original Draft

Yuanchang Xie: Conceptualization; Methodology; Writing – review & editing.

Helai Huang: Conceptualization; Writing – review & editing.

Pan Liu: Writing – review & editing.

Declaration of Competing Interest

The authors declare that there is no conflict of interest.

Acknowledgement

This research has been supported by the National Natural Science Foundation of China (71971061).

References

Alhajyaseen, W.K.M., 2015. The integration of conflict probability and severity for the safety assessment of intersections. Arab. J. Sci. Eng. 40 (2), 421–430.
Allen, B.L., Shin, B.T., Cooper, P., 1978. Analysis of traffic conflicts and collisions. Transp. Res. Rec. 667, 67–74.
Alonso, L., Pérez-Oria, J., 2010. Genetic Optimization of Fuzzy Adaptive Cruise Control for Urban Traffic. Fuzzy Modeling and Control: Theory and Applications 255–271.
Amundsen, F., Hyden, C., 1977. In: Proceedings of First Workshop on Traffic Conflicts. Oslo, Institute of Transport Economics.
Åsljung, D., Nilsson, J., Fredriksson, J., 2017. Using extreme value theory for vehicle level safety validation and implications for autonomous vehicles. IEEE Trans. Intell. Veh. 2 (4), 288–297.
Astarita, V., Giofré, V.P., 2019. From traffic conflict simulation to traffic crash simulation: introducing traffic safety indicators based on the explicit simulation of potential driver errors. Simul. Model. Pract. Theory. https://doi.org/10.1016/j.simpat.2019.03.003.
Astarita, V., et al., 2012. A new microsimulation model for the evaluation of traffic safety performances. Transp. Eur. 51, 1–16.
Autey, J., Sayed, T., Zaki, M., 2012. Safety evaluation of right-turn smart channels using automated traffic conflict analysis. Accid. Anal. Prev. 45, 120–130.
Bagdadi, O., 2013. Estimation of the severity of safety critical events. Accid. Anal. Prev. 50, 167–174.
Barcelo, J., Dumont, A.-G., Montero, L., Perarnau, J., Torday, A., 2003. Safety Indicators for Microsimulation-Based Assessments. Presented at 82nd Annual Meeting of the Transportation Research Board. Washington, D.C.
Blatt, A., Pierowicz, J., Flanigan, M., Lin, P.S., Kourtellis, A., Lee, C., Jovanis, P., Jenness, J., Wilaby, M., Campbell, J., Richard, C., 2015. Naturalistic Driving Study: Field Data Collection (No. SHRP 2 Report S2-S07-RW-1).
Boonsiripant, S., 2009. Speed Profile Variation As a Surrogate Measure of Road Safety Based on GPS-Equipped Vehicle Data (Doctoral Dissertation). Georgia Institute of Technology.
Cavadas, J., Azevedo, C.L., Farah, H., Ferreira, A., 2020. Road safety of passing maneuvers: a bivariate extreme value theory approach under non-stationary conditions. Accid. Anal. Prev. 134, 105315.
Chen, P., Zeng, W., Yu, G., Wang, Y., 2017. Surrogate safety analysis of pedestrian-vehicle conflict at intersections using unmanned aerial vehicle videos. J. Adv. Transp. 2017.
Cherian, M., Sathiyan, S., 2012. Neural Network based ACC for Optimized Safety and Comfort. International Journal of Computer Applications Volume 42– No.14 (0975-8887).
Chin, H.C., Quek, S.T., Cheu, R.L., 1992. Quantitative examination of traffic conflicts. Transp. Res. Rec. 1376, 86–91.
Cunto, F., 2008. Assessing Safety Performance of Transportation Systems Using Microscopic Simulation, Waterloo, Ontario, Canada.
Davis, G.A., Hourdos, J., Xiong, H., Chatterjee, I., 2011. Outline for a causal model of traffic conflicts and crashes. Accid. Anal. Prev. 43 (6), 1907–1919.
Dong, J., Li, L., Peng, H., Zhang, Y., 2020. A rule-based cooperative merging strategy for connected and automated vehicles. IEEE Trans. Intell. Transp. Syst.
Duan, S., Zhao, J., 2017. A model based on hierarchical safety distance algorithm for ACC control Mode switching strategy. 2017 2nd International Conference on Image, Vision and Computing.
Farah, H., Azevedo, C.L., 2017. Safety analysis of passing maneuvers using extreme value theory. IATSS Res. 41 (1), 12–21.
Federal Highway Administration, 2003. Surrogate Safety Measures From Traffic Simulation Models, Final Report. Publication No. FHWA-RD-03-050. Federal Highway Administration, USA.
Federal Highway Administration (FHWA), 2008. Surrogate Safety Assessment Model and Validation: Final Report. FHWA-HRT-08-051. Federal Highway Administration, USA.
Fernandes, P., Nunes, U., 2015. Multi-platooning leaders positioning and cooperative behavior algorithms of communicant automated vehicles for high traffic capacity. IEEE Trans. Intell. Transp. Syst. 16 (3), 1172–1187.
Flores, C., Milanes, V., 2018. Fractional-order-based ACC/CACC algorithm for improving string stability. Transp. Res. Part C Emerg. Technol. 95, 381–393.
Fraade-Blanar, L., Blumenthal, M.S., Anderson, J.M., Kalra, N., 2018. Measuring Automated Vehicle Safety: Forging a Framework.
Fu, T., Miranda-Moreno, L., Saunier, N., 2016. Pedestrian crosswalk safety at nonsignalized crossings during nighttime: use of thermal video data and surrogate safety measures. Transp. Res. Rec. 2586 (1), 90–99.
Fu, T., Wang, W., Li, Y., Xu, C.C., Xu, T., Li, X., 2019. Longitudinal safety impacts of cooperative adaptive cruise control vehicle's degradation. J. Safety Res. 69, 177–192.
Gay, K., Kniss, V., 2015. Safety Pilot Model Deployment: Lessons Learned and Recommendations for Future Connected Vehicle Activities (No. FHWA-JPO-16-363). United States. Department of Transportation. Intelligent Transportation Systems Joint Program Office.
Gettman, D., Head, L., 2003. Surrogate Safety Measures From Traffic Simulation Models (Report No. FHWA-RD-03-050).
Guo, F., Klauer, S.G., Hankey, J.M., Dingus, T.A., 2010. Near crashes as crash surrogate for naturalistic driving studies. Transp. Res. Rec. 2147 (1), 66–74.
Hayward, J.C., 1971. Near Misses As a Measure of Safety at Urban Intersections (Master Thesis).
He, Z., Qin, X., Liu, P., Sayed, M.A., 2018. Assessing surrogate safety measures using a safety pilot model deployment dataset. Transp. Res. Rec. 2672 (38), 1–11.
Huang, F., Liu, P., Yu Hao, Wang, W., 2013. Identifying if VISSIM simulation model and SSAM provide reasonable estimates for field measured traffic conflicts at signalized intersections. Accid. Anal. Prev. 50, 1014–1024.
Huang, K., Yang, X., Lu, Y., Mi, C.C., Kondlapudi, P., 2018. Ecological driving system for connected automated vehicles using a two-stage control hierarchy. IEEE Trans. Intell. Transp. Syst. 19 (7), 2373–2384.
Huang, K., Di, X., Du, Q., Chen, X., 2019. Stabilizing traffic via autonomous vehicles: a continuum mean field game approach. In: 2019 IEEE Intelligent Transportation Systems Conference (ITSC) Auckland. NZ, October 27-30, 2019.
Ishak, S.S., Osman, O.A., Codjoe, J., Jenkins, S., Karbalaieali, S., Theriot, M., Bakhit, P., 2017. Exploring Naturalistic Driving Data for Distracted Driving Measures (No. FHWA/LA. 17/580; Final Report 580). Louisiana Transportation Research Center.
Ismail, K., Sayed, T., Saunier, N., 2010. Automated analysis of pedestrian-vehicle conflicts: context for before-and-after Studies. Transp. Res. Rec. 2198, 52–64.
Jeong, E., Oh, C., Lee, S., 2017. Is vehicle automation enough to prevent crashes? Role of traffic operations in automated driving environments for traffic safety. Accid. Anal. Prev. 104, 115–124.
Jing, S., Hui, F., Zhao, X., Torres, J.R., 2019. Cooperative game approach to optimal merging sequence and on-ramp merging control of connected and automated vehicles. IEEE Trans. Intell. Transp. Syst. 20 (11), 4234–4244.
Johnsson, C., Laureshyn, A., De Ceunynck, T., 2018. In search of surrogate safety indicators for vulnerable road users: a review of surrogate safety indicators. Transp. Rev. 38 (6), 765–785.
Kanagaraj, V., Asaithambi, G., Toledo, T., Lee, T.C., 2015. Trajectory data and flow characteristics of mixed traffic. Transport. Res. Rec. J. Transport. Res. Board 2491, 1–11.
Kuang, Y., Qu, X., Wang, S., 2015. A tree-structured crash surrogate measure for freeways. Accid. Anal. Prev. 77, 137–148.
Laureshyn, A., Svensson, Å., Hydén, C., 2010. Evaluation of traffic safety, based on micro-level behavioural data: theoretical framework and first implementation. Accid. Anal. Prev. 42, 1637–1646.
Laureshyn, A., De Ceunynck, T., Karlsson, C., Svensson, Å., Daniels, S., 2017a. In search of the severity dimension of traffic events: extended Delta-V as a traffic conflict indicator. Accid. Anal. Prev. 98, 46–56.
Laureshyn, A., de Goede, M., Saunier, N., Fyhri, A., 2017b. Cross-comparison of three surrogate safety methods to diagnose cyclist safety problems at intersections in Norway. Accid. Anal. Prev. 105, 11–20.
Lee, J., Park, B., 2012. Development and evaluation of a cooperative vehicle intersection control algorithm under the connected vehicles environment. IEEE Trans. Intell. Transp. Syst. 13 (1), 81–90.
Li, Y., Li, Z., Wang, H., Wang, W., Xing, L., 2017a. Evaluating the safety impact of adaptive cruise control in traffic oscillations on freeways. Accid. Anal. Prev. 104, 137–145.
Li, Y., Wang, H., Wang, W., Xing, L., Liu, S., Wei, X., 2017b. Evaluation of the impacts of cooperative adaptive cruise control on reducing rear-end collision risks on freeways. Accid. Anal. Prev. 98, 87–95.
Li, Y., Xu, C., Xing, L., Wang, W., 2017c. Integrated cooperative adaptive cruise and variable speed limit controls for reducing rear-end collision risks near freeway bottlenecks based on micro-simulations. IEEE trans. Intell. Transp. Syst. 18 (11), 3157–3167.
Li, Y., Fu, T., Fan, Q., Dong, C., Wang, W., 2018. Influence of cyber-attacks on longitudinal safety of connected and automated vehicles. Accid. Anal. Prev. 121, 148–156.
Liu, X., Shen, D., Lai, L., Vine, S.L., 2020. Optimizing the safety-efficiency balancing of automated vehicle car following. Accid. Anal. Prev. 136, 105435.
Machiani, S.G., Abbas, M., 2016. Safety surrogate histograms (SSH): a novel real-time safety assessment of dilemma zone related conflicts at signalized intersections. Accid. Anal. Prev. 96, 361–370.

Malkhamah, S., Tight, M., Montgomery, F., 2005. The development of an automatic method of safety monitoring at Pelican crossings. Accident Analysis and Prevention 37, 938–946.
Milanes, V., Shladover, S.E., 2016. Handling cut-in vehicles in strings of cooperative adaptive cruise control vehicles. J. Intell. Transp. Syst. Technol. Plan. Oper. 20 (2), 178-191.
Minderhoud, M.M., Bovy, P.H.L., 2001. Extended time-to-collision measures for road traffic safety assessment. Accid. Anal. Prev. 33 (1), 89–97.
Mohtavipour, S.M., Mollajafari, M., 2019. An analytically derived reference signal to guarantee safety and comfort in adaptive cruise control systems. Journal of Intelligent transportation system. https://doi.org/10.1080/15472450.2019.1619559.
Morando, M.M., Tian, Q., Truong, L.T., Vu, H.L., 2018. Studying the safety impact of autonomous vehicles using simulation-based surrogate safety measures. J. Adv. Transp. 2018.
National Highway Traffic Safety Administration (NHTSA), 2016. Federal Automated Vehicles Policy.
Nister, D., Lee, H.L., Ng, J., Wang, Y., 2019. An Introduction to the Safety Force Field. NVDIA.
Oh, C., Park, S., Ritchie, S.G., 2006. A method for identifying rear-end collision risks using inductive loop detectors. Accid. Anal. Prev. 38, 295–301.
Okamura, M., Fukuda, A., Morita, H., Suzuki, H., Nakazawa, M., 2011. Impact evaluation of a driving support system on traffic flow by microscopic traffic simulation. Adv. Transp. Stud.: Int. J. (Special Issue 2011).
Orsini, F., Gecchele, G., Gastaldi, M., Rossi, R., 2019. Collision prediction in roundabouts: a comparative study of extreme value theory approaches. Transp. A Transp. Sci. 15 (2), 556–572.
Orsini, F., Gecchele, G., Gastaldi, M., Rossi, R., 2020. Large-scale road safety evaluation using extreme value theory. IET Intell. Transp. Syst.
Ozbay, K., Yang, H., Bartin, B., Mudigonda, S., 2008. Derivation and validation of new simulation-based surrogate safety measure. Transp. Res. Rec. 105–113.
Papadoulis, A., Quddus, M., Imprialou, M., 2019. Evaluating the safety impact of connected and autonomous vehicles on motorways. Accid. Anal. Prev. 124, 12–22.
Park, H., Oh, C., 2019. A vehicle speed harmonization strategy for minimizing inter-vehicle crash risks. Accid. Anal. Prev. 128, 230–239.
Perkins, S.R., Harris, J.I., 1967. Traffic conflict characteristics: Freeway curve and exit area F1. December, 1966. http://trid.trb.org/view/618537. Retrieved from:
Qu, X., Yu, Y., Zhou, M., Lin, C.T., Wang, X., 2020. Jointly dampening traffic oscillations and improving energy consumption with electric, connected and automated vehicles: a reinforcement learning based approach. Appl. Energy 257, 114030.
Rahman, M.S., Abdel-Aty, M., 2018. Longitudinal safety evaluation of connected vehicles' platooning on expressways. Accid. Anal. Prev. 117, 381–391.
Rahman, M.S., Abdel-Aty, M., Wang, L., Lee, J., 2018. Understanding the highway safety benefits of diff ;erent approaches of connected vehicles in reduced visibility conditions. Presented at the Transportation Research Board (TRB) 97th Annual Meeting.
Rahman, M.S., Abdel-Aty, M., Lee, J., 2019a. Understanding the safety benefits of connected and automated vehicles on arterials' intersections and segments. In: Presented at 98th Annual Meeting of Transportation Research Board. Washington, D. C..
Rahman, M.S., Abdel-Aty, M., Lee, J., Rahman, M.H., 2019b. Safety benefits of arterials' crash risk under connected and automated vehicles. Transp. Res. Part C Emerg. Technol. 100, 354–371.
Ren, T., Xie, Y., Jiang, L., 2020. Cooperative highway work zone merge control based on reinforcement learning in a connected and automated environment. Transp. Res. Record 2674, 363–374.
Ren, T., Xie, Y., Jiang, L., 2021. New England merge: a novel cooperative merge control method for improving highway work zone mobility and safety. J. Intell. Transp. Syst. Technol. Plan. Oper. 25 (1), 107–121.
Roncoli, C., Papamichail, I., Papageorgiou, M., 2015. Model predictive control for motorway traffic with mixed manual and VACS-equipped vehicles. In: 18th Euro Working Group on Transportation, EWGT 2015. 14-16 July 2015, Delft, the Netherlands.
Saunier, N., Sayed, T., 2008. Probabilistic framework for automated analysis of exposure to road collisions. J. Transp. Res. Record 2083, 96–104.
Shalev-Shwartz, S., Shammah, S., Shashua, A., 2018. Vision Zero: Can Roadway Accidents Be Eliminated Without Compromising Traffic Throughput.
Shelby, S.G., 2011. Delta-v as a measure of traffic conflict severity. In: 90th TRB Annual Meeting. Washington, DC..
Songchitruksa, P., Tarko, A., 2006. The extreme value theory approach to safety estimation. Accid. Anal. Prev. 38, 811–822.
Stipancic, J., Miranda-Moreno, L., Saunier, N., 2018. Vehicle maneuvers as surrogate safety measures: extracting data from the GPS-enabled smartphones of regular drivers. Accid. Anal. Prev. 115, 160–169.
Strauss, J., Zangenehpour, S., Miranda-Moreno, L.F., Saunier, N., 2017a. Cyclist deceleration rate as surrogate safety measure in Montreal using smartphone GPS data. Accid. Anal. Prev. 99, 287–296.
Strauss, J., Zangenehpour, S., Miranda-Moreno, L.F., Saunier, N., 2017b. Cyclist deceleration rate as surrogate safety measure in Montreal using smartphone GPS data. Accid. Anal. Prev. 99, 287–296.
Tarko, A.P., 2018. Estimating the expected number of crashes with traffic conflicts and the Lomax Distribution – a theoretical and numerical exploration. Accid. Anal. Prev. 113, 63–73.
Tarko, A.P., 2020. Chapter 3 - traffic conflicts as crash surrogates. In: Tarko, A.P. (Ed.), Measuring Road Safety Using Surrogate Events. Elsevier, pp. 31–45.
Tarko, A., Davis, G., Saunier, N., Sayed, T., Washinton, S., 2009. White paper: surrogate measures of safety. Committee on Safety Data Evaluation and Analysis (ANB20).
Tibljaš, A.D., Giuffre, T., Surdonja, S., Trubia, S., 2018. Introduction of autonomous vehicles: roundabouts design and safety performance evaluation. Sustainability 10, 1060.
Topp, H.H. (Ed.), 1998. Traffic safety work with video-processing. University Kaiserslautern. Transportation Department. Green Series No.43.
Torres, J.R., Malikopoulos, A., 2017. Automated and cooperative vehicle merging at highway on-ramps. IEEE Trans. Intell. Transp. Syst. 18 (4), 780–789.
Uno, N., et al., 2002. A microscopic analysis of traffic conflict caused by lane-changing vehicle at weaving section. Proceedings of the 13th Mini-Euro Conference Handling Uncertainty in Transportation Analysis of Traffic and Transportation Systems.
USDOT Intelligent Transportation Systems (ITS) - Joint Program Office, 2021. Connected Vehicle Pilot (CVP) Open Data. Available online athttps://data.transportation.gov/stories/s/Connected-Vehicle-Pilot-Sandbox/hr8h-ufhqAccessed on Feb 24, 2021.
Várhelyi, A., 1998. Driver's speed behavior at a zebra crossing: a case study. Accid. Anal. Prev. 30 (6), 731–743.
Virdi, N., Grzybowska, H., Waller, S.T., Dixit, V., 2019. A safety assessment of mixed fleets with connected and Autonomous Vehicles using the Surrogate Safety Assessment Module. Accid. Anal. Prev. 131, 95–111.
Vogel, K., 2003. A comparison of headway and time to collision as safety indicators. Accid. Anal. Prev. 37, 263–295.
Wang, M., 2018. Infrastructure assisted adaptive driving to stabilise heterogeneous vehicle strings. Transp. Res. Part C Emerg. Technol. 91, 276–295.
Wang, C., Stamatiadis, N., 2013. Surrogate Safety Measure for Simulation-Based Conflict Study. In Transportation Research Record: Journal of the Transportation Research Board, No. 2386. Transportation Research Board of the National Academies, Washington, D.C, pp. 72–80.
Wang, C., Stamatiadis, N., 2014a. Derivation of a new surrogate measure of crash severity. Transp. Res. Rec. 2014 (2432), 37–45.
Wang, C., Stamatiadis, N., 2014b. Evaluation of a simulation-based surrogate safety metric. Accid. Anal. Prev. 71, 82–92.
Wang, C., Stamatiadis, N., 2016. Sensitivity analysis on new simulation-based conflict metrics. Saf. Sci. 82, 399–409.
Wang, C., Xu, C., Xia, J., Qian, Z., 2018. A combined use of microscopic traffic simulation and extreme value methods for traffic safety evaluation. Transportation Research Part C, Vol. 90, 281–291.
Wang, C., Xu, C., Dai, Y., 2019a. A crash prediction method based on bivariate extreme value theory and video-based vehicle trajectory data. Accid. Anal. Prev. 123, 365–373.
Wang, M., Maarseveen, S.V., Happee, R., Tool, O., Arem, B.V., 2019b. Benefits and risks of truck platooning on freeway operations near entrance ramp. Transp. Res. Rec. 2673 (8), 588–602.
Weng, B., Rao, S., Deosthale, E., Schnelle, S., Barickman, F., 2020. Model predictive instantaneous safety metric for evaluation of automated driving systems. Accepted at IEEE Intelligent Vehicles Symposium (IV), 2020.
Winner, H., Lemmer, K., Form, T., Mazzega, J., 2019. Pegasus—first steps for the safe introduction of automated driving. Road Vehicle Automation 5. Springer, Cham, pp. 185–195.
Wu, J., Xu, H., Zheng, Y., Tian, Z., 2018. A novel method of vehicle-pedestrian near-crash identification with roadside LiDAR data. Accid. Anal. Prev. 121, 238–249.
Wu, Y., Abdel-Aty, M., Wang, L., Rahman, M.S., 2019. Combined connected vehicles and variable speed limit strategies to reduce rear-end crash risk under fog conditions. Journal of Intelligent Transportation 2019. https://doi.org/10.1080/15472450.2019.1634560.
Xie, Y., Gartner, N.H., Stamatiadis, P., Ren, T., Salcedo, G., 2018. Optimizing Future Work Zones in New England for Improved Safety and Mobility (No. NETC 14-4).
Xie, K., Yang, D., Ozbay, K., Yang, H., 2019. Use of real-world connected vehicle data in identifying high-risk locations based on a new surrogate safety measure. Accid. Anal. Prev. 125, 311–319.
Xu, X., Grizzle, J.W., Tabuada, P., Ames, A.D., 2018. Correctness guarantees for the composition of lane keeping and adaptive cruise control. IEEE Trans. Autom. Sci. Eng. 15 (3), 1216–1229.
Xu, H., Feng, S., Zhang, Y., Li, L., 2019. A grouping-based cooperative driving strategy for CAVs merging problems. IEEE Trans. Veh. Technol. 68 (6), 6125–6136.
Xu, H., Zhang, Y., Li, L., Li, W., 2020. Cooperative driving at unsignalized intersections using tree search. IEEE Trans. Intell. Transp. Syst. https://doi.org/10.1109/TITS.2019.2940641.
Yao, S., Friedrich, B., 2019. Managing connected and automated vehicles in mixed traffic by human-leading platooning strategy: a simulation study. In: 2019 IEEE Intelligent Transportation Systems Conference (ITSC). Auckland, NZ, October 27-30.
Zhao, R.C., Wong, P.K., Xie, Z.C., Zhao, J., 2017. Real-time weighted multi-objective model predictive controller for adaptive cruise control systems. Int. J. Automot. Technol. 18 (2), 279–292.
Zheng, L., Sayed, T., 2019. From univariate to bivariate extreme value models: approaches to integrate traffic conflict indicators for crash estimation. Transp. Res. Part C Emerg. Technol. 103, 211–225.
Zheng, L., Ismail, K., Meng, X., 2014a. Freeway safety estimation using extreme value theory approaches: a comparative study. Accid. Anal. Prev. 62, 32–41.
Zheng, L., Ismail, K., Meng, X., 2014b. Shifted gamma-generalized pareto distribution model to map the safety continuum and estimate crashes. Saf. Sci. 64, 155–162.
Zheng, L., Ismail, K., Sayed, T., Fatema, T., 2018. Bivariate extreme value modeling for road safety estimation. Accid. Anal. Prev. 120, 83–91.
Zheng, L., Sayed, T., Essa, M., 2019a. Validating the bivariate extreme value modeling approach for road safety estimation with different traffic conflict indicators. Accid. Anal. Prev. 123, 314–323.

Zheng, L., Sayed, T., Essa, M., Guo, Y., 2019b. Do simulated traffic conflicts predict crashes? An investigation using the extreme value approach. In: 2019 IEEE Intelligent Transportation Systems Conference. ITSC 2019.

Zhong, Z., Nejad, M., Lee, E.E., Lee, J., 2019. Clustering strategies of cooperative adaptive cruise control: impacts on human-driven vehicles. 2019 IEEE 2nd Connected and Automated Vehicles Symposium.

Zhou, M., Yu, Y., Qu, X., 2020. Development of an efficient driving strategy for connected and automated vehicles at signalized intersections: a reinforcement learning approach. IEEE Trans. Intell. Transp. Syst. 21 (1), 433–443.